

山东省联合农药工业有限公司

SHANDONG UNITED PESTICIDE INDUSTRY CO., LTD.

ADD: Building 1#, Middle Shengli Road, Daxin Village, Fan Town, Daiyue District, Taian City, Shandong, P.R.China.

TEL: 0086-0531-88977085 FAX: 0086-0531-86401690

MATERIAL SAFETY DATA SHEET

PROFENOFOS 400g/l · CYPERMETHRIN 40g/l EC

1. Chemical Product Identification

1.1 Product Details

Product Name: Profenofos 400g/l · Cypermethrin 40g/l EC

For Profenofos TC

CAS RN: 41198-08-7

Molecular Formula: $C_{11}H_{15}BrClO_3PS$

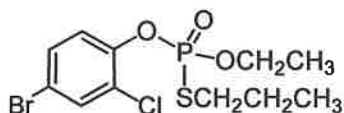
Molecular Weight: 373.63

Chemical Name:

IUPAC: (RS)-(O-4-bromo-2-chlorophenyl O-ethyl S-propyl phosphorothioate)

CAS: O-(4-bromo-2-chlorophenyl) O-ethyl S-propyl phosphorothioate

Structure:



For Cypermethrin TC

CAS RN: 52315-07-8

Molecular Formula: $C_{22}H_{19}Cl_2NO_3$

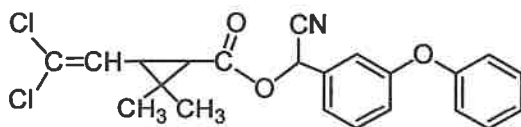
Molecular Weight: 416.32

Chemical Name:

IUPAC: (RS)- α -cyano-3-phenoxybenzyl (1RS, 3RS; 1RS, 3SR) -3 - (2, 2 - dichlorovinyl) -2, 2 -dimethylcyclopropanecarboxylate

CAS:cyano(3-phenoxyphenyl)methyl-3-(2,2-dichloroethenyl)-2,2-dimethylcyclopropanecarboxylate

Structure:



1.2 Company Identification

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Name: SHANDONG UNITED PESTICIDE INDUSTRY CO., LTD

Address: Building 1#, Middle Shengli Road, Daxin Village, Fan Town, Daiyue District City, People's Republic of China.

Telephone number: 86-531-86401640

2. Composition / Information on Ingredients

Compositio	CAS No.	Content, g/l
Profenofos	41198-08-7	400Min.
Cypermethrin	52315-07-8	40 Min.
Others	-	Up to 1L

3. Hazards Identification

Health Hazards: Harmful if swallowed. Irritating to skin. May cause sensitization by skin contact. Risk of serious damage to eyes.

Environmental hazard in case of accident (Spillage/Leakage): Very toxic to aquatic organisms.

4. First Aid Measures

First-Aid Measures

Ensure that adequate decontamination has been carried out. If patient is not breathing, start artificial respiration, preferably with a demand-valve resuscitator, bag-valve-mask device, or pocket mask, as trained. Perform CPR as necessary. Immediately flush contaminated eyes with gently flowing water. Do not induce vomiting. If vomiting occurs, lean patient forward or place on left side (head-down position, if possible) to maintain an open airway and prevent aspiration. Keep patient quiet and maintain normal body temperature. Obtain medical attention.
/Pyrethrins, pyrethroids, and related compounds/

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5. Fire-Fighting Measures

Extinguishing media

Suitable Extinguishing Media: Dry chemical extinguisher, foam, carbon dioxide or waterspray (do not use direct jet of water).

Fire fighting

Special Hazards during Fire Fighting: Combustion products are toxic and/or irritant. Measures have to be taken to prevent the contaminated extinguishing agent from seeping into the ground or from spreading uncontrollably.

Protective Equipment for Fire Fighting: Use self contained breathing apparatus. Wear protective equipment.

6. Accidental Release Measures

Environmental Protection Measures following Accidents: Soak up with absorptive material such as sand, soil, diatomaceous earth, etc. Prevent material from spreading, e.g. by damming in with absorptive material. Collect material in specially marked, tightly closing containers. Spilled product cannot be used further and must be disposed of. If safe disposal is not possible, contact the manufacturer, the dealer or the local representative. Do not contaminate waters and sewers. Keep away from sources of ignition.

7. Handling and Storage

Handling information

General: Avoid contact with skin, eyes and clothing. Avoid inhalation of fog and vapours. Keep away from sources of ignition. Take precautionary measures against static discharges (transfer operations: check grounding). May form explosive vapour-air mixtures. Do not eat, drink or smoke while working. In addition to the measures usually taken in chemical works like splashproof filling and measuring equipment (including vapour stripping), further personal protection measures may

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have to be implemented to avoid possible contact with the product.

Storage information

Special Instructions for Storage: Store the product in closed original containers.

Protect from light and humidity. Storage depot protected against fire.

Storage Compatibility with other Products: Store separate from food and feed.

8. Exposure Controls / Personal Protection

Personal protective equipment

In General: Change working clothes daily.

Breathing Protection: In case of heavy exposure, wear: Gas mask with universal filter.

Eye Protection: Goggles or face-shield

Hand Protection: Chemical-resistant gloves

Body Protection: Heavy duty cotton or synthetic fabric working clothes (e.g. overalls). Heavy-duty shoes or boots.

Precautionary measures after work: Wash thoroughly (shower, bathe, wash hair). Change clothing. Thoroughly clean protective gear. Thoroughly clean contaminated equipment with soap or soda solution.

9. Physical and Chemical Properties

Form: liquid

Color: light brown to brown

Physical State: liquid

Emulsion stability (Dilute 200): Stable

10. Stability and Reactivity

Chemical Stability

Assessment: Stable under standard conditions.

Corrosiveness

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Corrosiveness / Appropriate Material: tin-plate

11. Toxicological Information

Acute Oral Toxicity

LD₅₀: 300-2000 mg/kg (rat)

Acute Dermal Toxicity

LD₅₀: > 2000 mg/kg (rat)

Acute Inhalation Toxicity

LC₅₀: 2.84 mg/m³ (rat; 4 h)

Skin Irritation: Non-irritant (rabbit)

Eye Irritation: Irritating to eyes (rabbit)

Skin Sensitization: May be a weak skin sensitiser.

12. Ecological and Ecotoxicological Information

For Profenofos:

Birds LC₅₀ (8 d) for bobwhite quail 70-200, Japanese quail >1000, mallard ducks 150-612 ppm.

Fish LC₅₀ (96 h) for rainbow trout 0.08, crucian carp 0.09, bluegill sunfish 0.3 mg/l.

Daphnia EC₅₀ (48 h) 1.06 µg/l.

Algae EC₅₀ (72 h) for *Scenedesmus subspicatus* 1.16 mg/l.

Bees LD₅₀ (contact, 48 h) 0.102 µg/bee.

Worms LC₅₀ (14 d) 372 mg/kg.

For Cypermethrin:

Birds Acute oral LD₅₀ for mallard ducks >10 000, chickens >2000 mg/kg.

Fish LC₅₀ (96 h) for rainbow trout 0.69, sheepshead minnow 2.37 µg/l; under field conditions, fish are not at risk from normal agricultural usage.

Daphnia LC₅₀ (48 h) 0.15 µg/l.

Bees Highly toxic to honeybees in laboratory tests, but field applications at recommended dosages do not put hives at risk. LD₅₀ (24 h) (oral) 0.035 µg/bee;

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(topical) 0.02 µg/bee.

13. Disposal Considerations

Disposal

Dispose of surplus product on a landfill site approved for pesticides or bury in a safe place away from water supplies. Dilute surplus application mixture and pour it broadcast on fallow land. Empty metal containers are to be flattened, perforated and disposed of on a landfill site or buried in a safe place. Dispose of plastic containers on a landfill site or burn at a safe place. For drums take re-cycling in account, otherwise crush and bury. Do not re-use empty containers for any other purpose, especially not for storing drinking water, food or feed.

14. Transport Information

UN-Number: 3082;

Hazard Class: 9

Packaging group: III

15. Regulatory Information

Hazard Symbol Xn Harmful

Special Risks

Flammable.

Harmful if swallowed.

Irritating to skin.

Risk of serious damage to eyes.

May cause sensitization by skin contact.

16. Other Information

All information and instructions provided in this Material Safety Data Sheet (MSDS) are based on the current state of scientific and technical knowledge at the date indicated on the present MSDS and are presented in good faith and believed to be

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correct. This information applies to the product as such. In case of new formulations or mixes, it is necessary to ascertain that a new danger will not appear. It is the responsibility of persons on receipt of this MSDS to ensure that the information contained herein is properly read and understood by all people who may use, handle, dispose or in any way come in contact with the product. If the recipient subsequently produce formulations containing this product, it is the recipients sole responsibility to ensure the transfer of all relevant information from this MSDS to their own MSDS.